

$$z_1 = w_1 x_1 + b_1$$

$$A_1 = \text{Relu}(z_1)$$

$$z_2 = w_2 A_1 + b_2$$

$$A_2 = \text{Relu}(z_2)$$

$$z_3 = w_3 A_2 + b_3$$

$$y^{\wedge} = \text{Softmax}(z_3)$$

$$C = - \sum y (\log(y^{\wedge})) \leftarrow \text{cross entropy}$$

$$\frac{\partial C}{\partial y^{\wedge}} = - \sum y \frac{1}{y^{\wedge}} \leftarrow \text{sw}$$

$$\frac{\partial y^{\wedge}}{\partial z_3} = \text{Softmax}'(z_3)$$

$$\text{Softmax}'(z) = \begin{matrix} e_i \\ \sum e_i \end{matrix}$$

$$\frac{\partial C}{\partial z_3} = y^{\wedge} - y$$

$$\frac{\partial C}{\partial B_3} = \frac{\partial C}{\partial z_3}$$

$$\frac{\partial C}{\partial w_3} = \frac{\partial C}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_3}$$

$$\frac{\partial z_3}{\partial w_3} = \frac{\partial w_3}{\partial w_3} A_2 + 0 = A_2$$

$$\frac{\partial C}{\partial w_3} = (y^{\wedge} - y) \cdot A_2$$

$$\frac{\partial C}{\partial b_3} = (y^{\wedge} - y)$$

$$\frac{\partial C}{\partial w_2} = \frac{\partial C}{\partial z_3} \cdot \frac{\partial z_3}{\partial A_2} \cdot \frac{\partial A_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_2}$$

$$\frac{\partial C}{\partial w_2} = (y^{\wedge} - y) \cdot w_3 \cdot \text{Relu}'(z_2) \cdot A_3$$

$$\frac{\partial z_2}{\partial w_2} = \frac{\partial w_2}{\partial w_2} A_1 + b_2 = A_1$$

$$\frac{\partial C}{\partial z_2} = (y^{\wedge} - y) \cdot w_3 \cdot \text{Relu}'(z_2)$$

$$\frac{\partial C}{\partial w_2} = (y^{\wedge} - y) \cdot w_3 \cdot \text{Relu}'(z_2) \cdot A_1 = \frac{\partial C}{\partial z_2} \cdot A_1$$

$$\frac{\partial C}{\partial b_2} = (y^{\wedge} - y) \cdot w_3 \cdot \text{Relu}'(z_2) = \frac{\partial C}{\partial z_2}$$

$$\frac{\partial C}{\partial z_2} = (y^{\wedge} - y) \cdot w_3 \cdot \text{Relu}'(z_2)$$

$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial z_2} \cdot \frac{\partial z_2}{\partial A_1} \cdot \frac{\partial A_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

$$\therefore \frac{\partial z_1}{\partial w_1} = 1, \quad \frac{\partial z_1}{\partial b_1} = 1$$

$$\frac{\partial z_2}{\partial z_1} = \frac{\partial C}{\partial z_2} \cdot w_2 \cdot \text{Relu}'(z_1)$$

$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial z_1} \cdot x, \quad \frac{\partial C}{\partial b_1} = \frac{\partial C}{\partial z_1}$$

$$z_1 = [w_1] [x_1] + [B_1] \rightarrow 16 \times 1$$

$16 \times 10 \quad 10 \times 1 \quad 16 \times 1$
 $A_1 = \text{Relu}(z_1)$

$$z_2 = [w_2] [A_1] + [B_2] \rightarrow 16 \times 1$$

$16 \times 16 \quad 16 \times 1 \quad 16 \times 1$
 $A_2 = \text{Relu}(z_2)$

$$z_3 = [w_3] [A_2] + [B_3] \rightarrow 3 \times 1$$

$3 \times 16 \quad 16 \times 1 \quad 3 \times 1$
 $y^{\wedge} = \text{Softmax}(z_3)$

$$B_2 \rightarrow 16 \times 1$$

$$\frac{\partial C}{\partial B_2} = (y^{\wedge} - y) \cdot w_3 \cdot \text{Relu}'(z_2)$$

$3 \times 1 \quad 3 \times 16 \quad 16 \times 1$
 $= [w_3.T] [y^{\wedge} - y] \cdot \text{Relu}'(z_2)$
 $16 \times 3 \quad 3 \times 1 \quad 16 \times 1$

$$16 \times 1 \cdot z \cdot 16 \times 1 \leftarrow \partial B_2$$

$$w_2 \rightarrow 16 \times 16$$

$$\frac{\partial C}{\partial w_2} = \frac{\partial C}{\partial B_2} \cdot A_1$$

$16 \times 1 \quad 16 \times 1$

$$= [\partial B_2] \cdot [A_1.T]$$

$16 \times 1 \quad 1 \times 16$

$$16 \times 16 \cdot z \cdot 16 \times 16 \checkmark$$

$$b_1 \rightarrow 16 \times 1$$

$$db_1 = dz_2 \cdot w_2 \cdot \text{Relu}'(z_1)$$

$$db_2, w_2 \cdot \text{Relu}'(z_1)$$

$16 \times 1 \quad 16 \times 16 \quad 16 \times 1$

$$= [w_2^T] [db_2] \cdot \text{Relu}'(z_1)$$

$16 \times 16 \quad 16 \times 1$

$$16 \times 1 = 16 \times 1 \leftarrow db_1$$

$$w_1 \rightarrow 16 \times 10$$

$$dw_1 = db_1 \cdot X$$

$16 \times 1 \quad 16 \times 1$

$$= [db_1] [X^T]$$

$16 \times 1 \quad 1 \times 10$

$$dw_1 = 16 \times 10 \checkmark$$